Customer Behaviour Analysis Report

InternSpark Data Analyst Internship Project

Dataset: Consumer Shopping Trends (Kaggle)

Tools: Python | Pandas | Matplotlib | Seaborn | Google Colab

April 2026

This report presents a comprehensive end-to-end data analysis of e-commerce customer shopping behaviour. It covers data cleaning, exploratory analysis, 10 key visualizations, findings, and actionable business recommendations.

1. Introduction

Customer Behaviour Analysis is the process of examining and understanding how customers interact with a business, its products, and services. By analysing purchasing patterns, preferences, and demographics, businesses can make data-driven decisions to improve customer experiences, optimise marketing strategies, and maximise revenue.

This project was completed as part of the InternSpark Data Analyst Internship. The objective was to perform an end-to-end analysis of a real-world e-commerce dataset, answer key business questions, and produce actionable insights.

Business Questions Addressed

1. How much do customers typically spend per transaction?
2. Which gender makes more purchases and spends more on average?
3. What are the most popular product categories?
4. Which age group spends the most money?
5. What is the most preferred payment method?
6. In which season do customers spend the most?
7. Which product category generates the highest average revenue?

8. Do subscribed customers spend more than non-subscribed customers?
9. What are the top 10 most purchased individual items?
10. Which numerical factors are correlated with each other?

2. Dataset Overview

The dataset used in this project is the **Consumer Shopping Trends Dataset** sourced from Kaggle. It contains transactional data from an e-commerce platform capturing various attributes of customer purchases.

Source	Kaggle — Consumer Shopping Trends Dataset
File	shopping_trends_updated.csv
Total Records	3,900 customer transactions
Total Columns	18 features
Data Types	Numerical (Age, Purchase Amount, Review Rating) and Categorical (Gender, Category, Season, Payment Method)

Key Columns in the Dataset

Column Name	Description	Type
Customer ID	Unique identifier for each customer	Numerical
Age	Age of the customer	Numerical
Gender	Male or Female	Categorical
Item Purchased	Name of the product purchased	Categorical
Category	Product category (Clothing, Electronics, etc.)	Categorical
Purchase Amount (USD)	Amount spent in US Dollars	Numerical
Season	Season when purchase was made	Categorical
Payment Method	Mode of payment used	Categorical
Subscription Status	Whether customer is subscribed	Categorical
Review Rating	Customer rating (1-5)	Numerical

3. Data Cleaning

Before performing analysis, the dataset was thoroughly cleaned to ensure accuracy and reliability of results. The following steps were performed using Python and the Pandas library:

✓ Missing Values Check

Used df.isnull().sum() to identify missing values across all columns. Result: No missing values were found in the dataset.

✓ Duplicate Rows Check

Used df.duplicated().sum() to check for duplicate entries. Result: 0 duplicate rows found — dataset was already clean.

✓ Data Type Verification

Verified that numerical columns (Age, Purchase Amount) were correctly typed as int64/float64, and categorical columns were object type.

✓ Age Group Creation

Created a new derived column 'Age Group' by binning the Age column into meaningful groups: Under 18, 18-25, 26-35, 36-50, and 50+.

Final Clean Dataset: 3,900 rows × 18 columns — Ready for Analysis

4. Analysis and Key Findings

Ten visualizations were created to answer the business questions. Each chart was generated using Matplotlib and Seaborn in Google Colab and saved as high-resolution PNG files.

Chart 1 — Purchase Amount Distribution

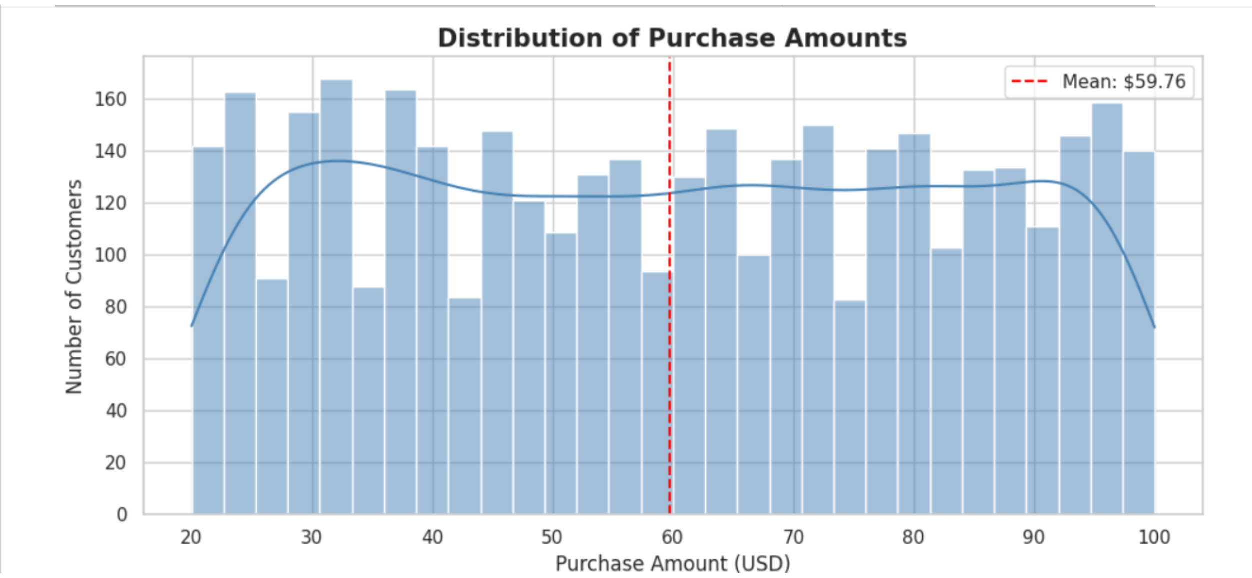


Chart 2 — Purchases by Gender

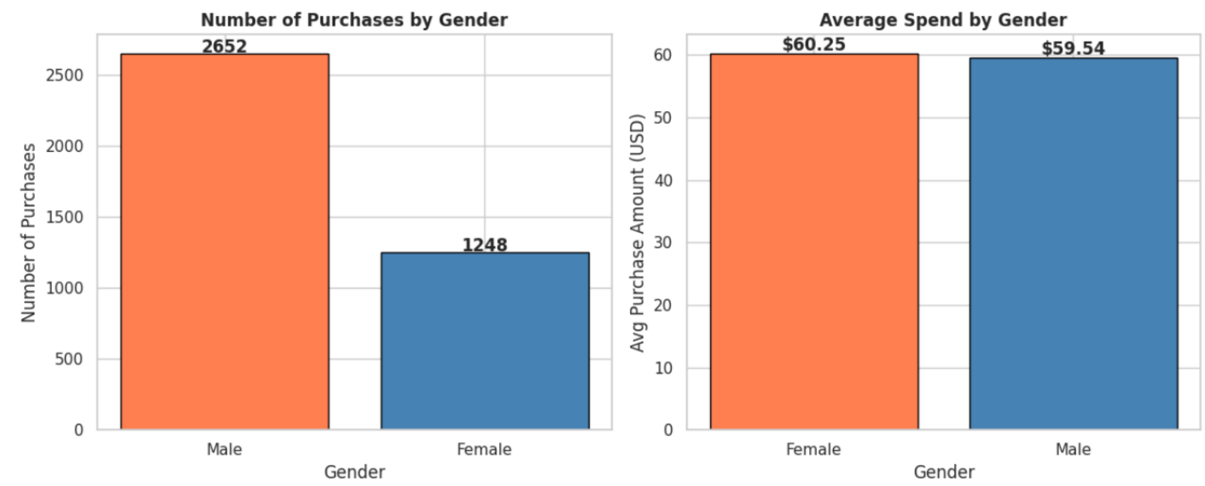


Chart 3 — Most Popular Product Categories

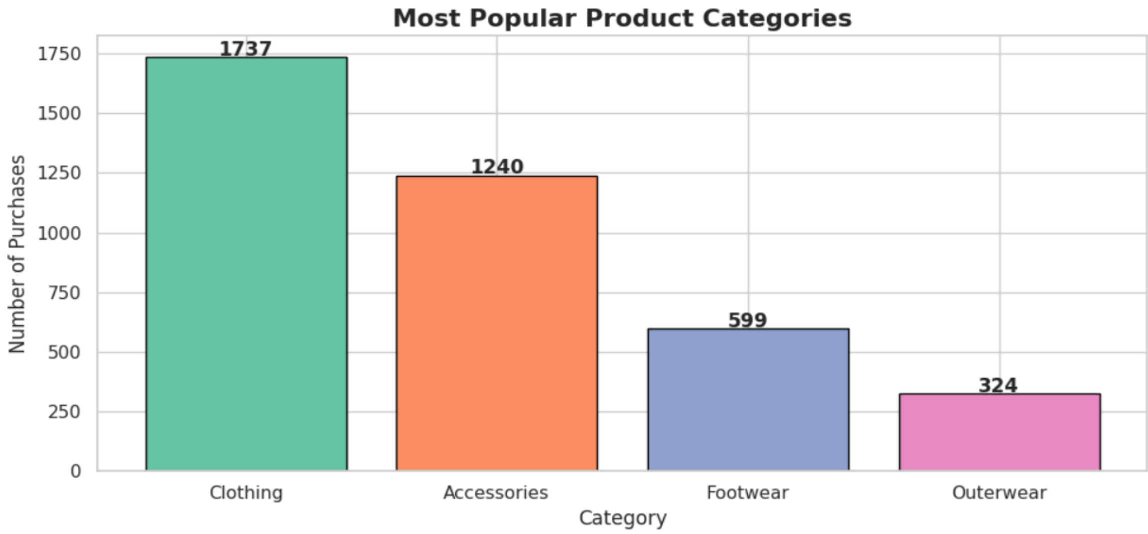
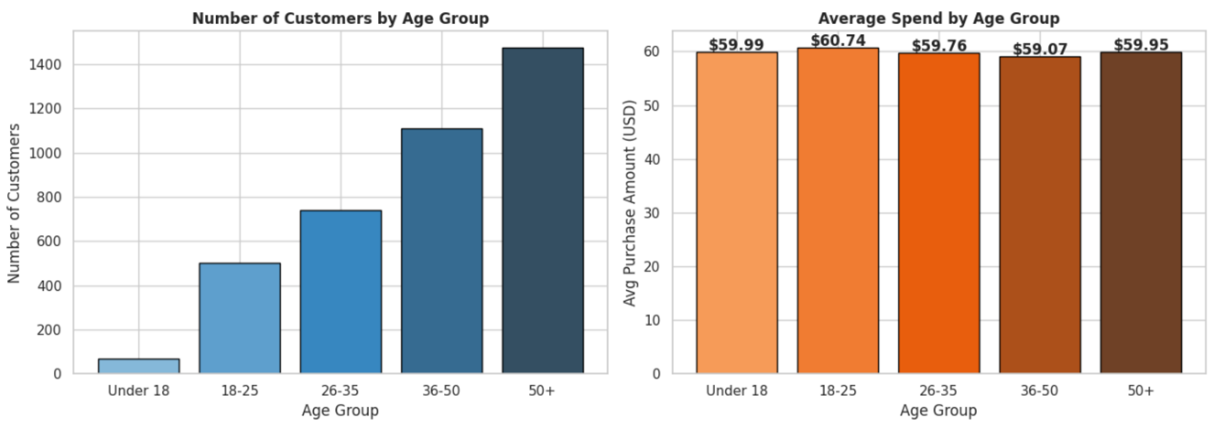


Chart 4 — Purchase by Age Group



📌 Insight: The 18-25 age group has the highest average spend at $60.74

Chart 5 — Preferred Payment Methods

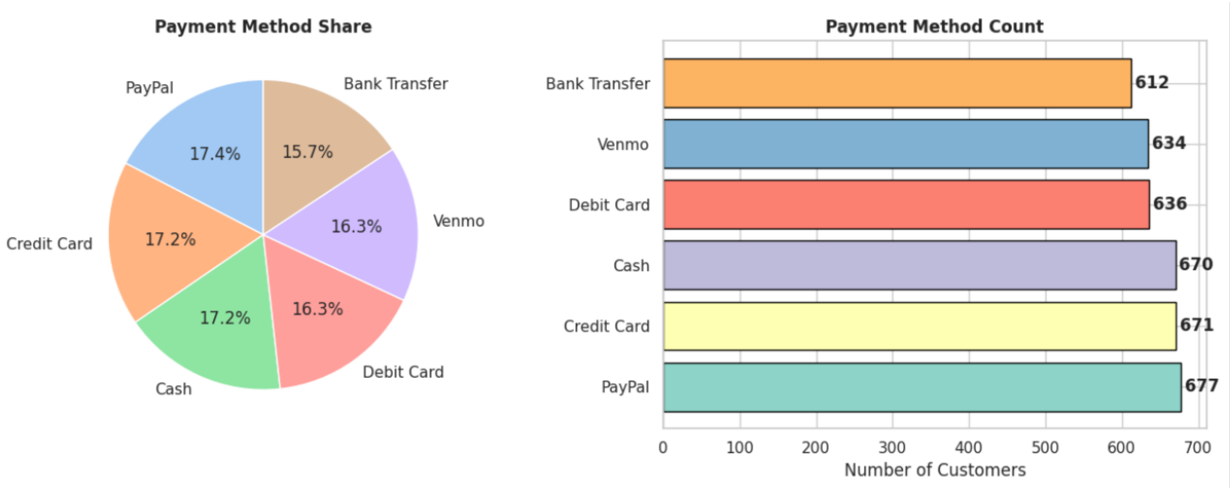


Chart 6 — Seasonal Purchasing Trends

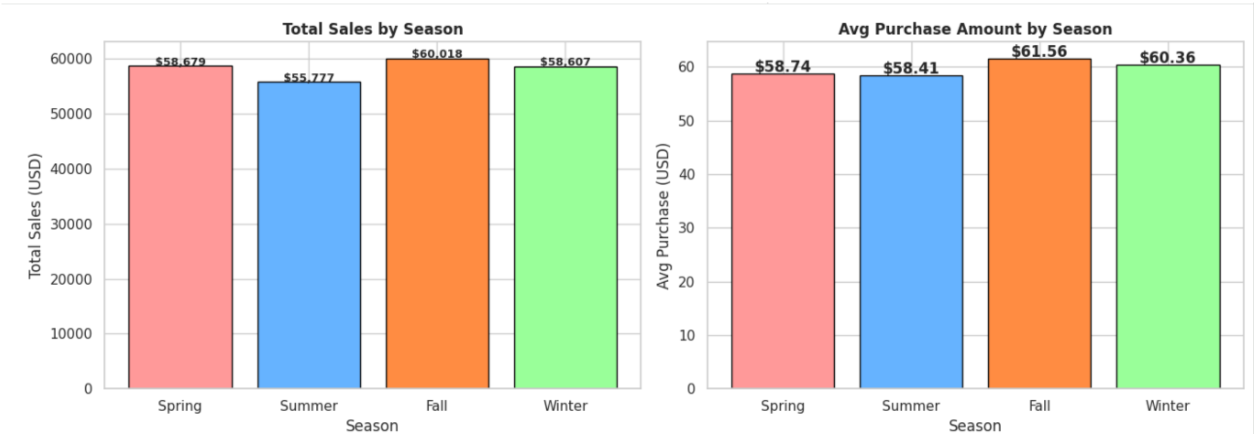


Chart 7 — Category vs Average Purchase Amount

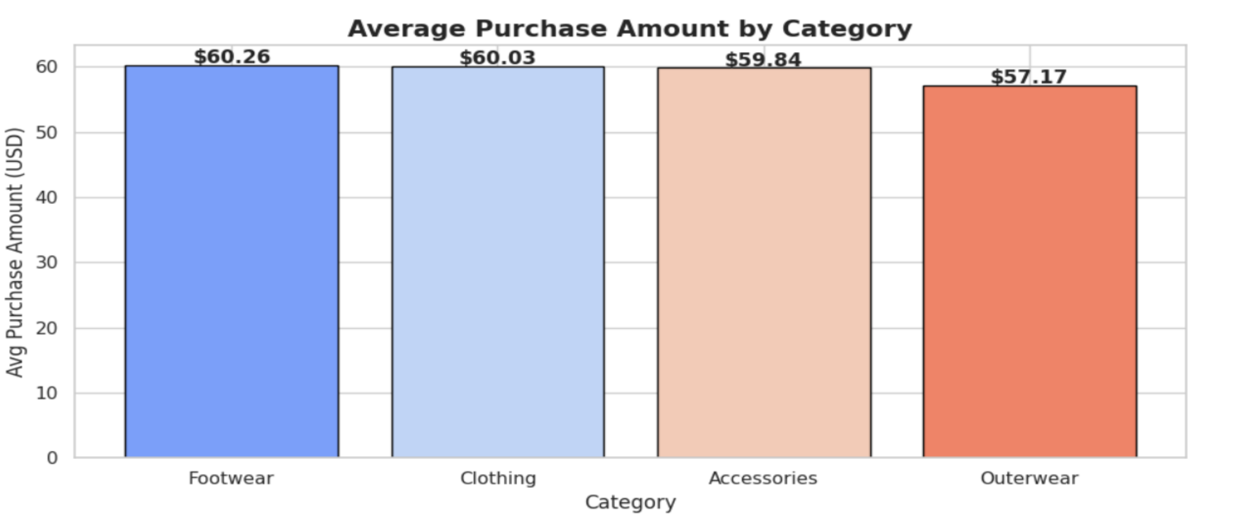


Chart 8 — Correlation Heatmap

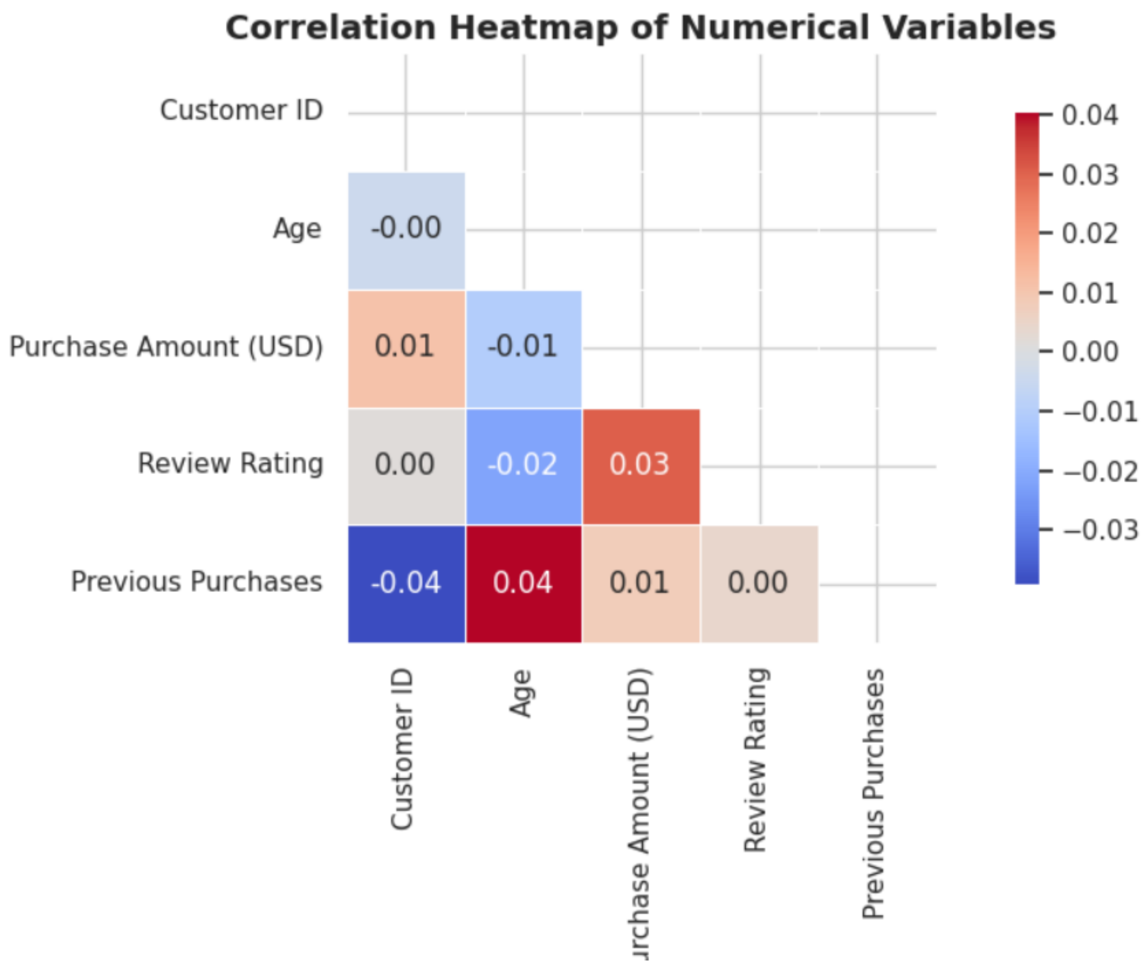
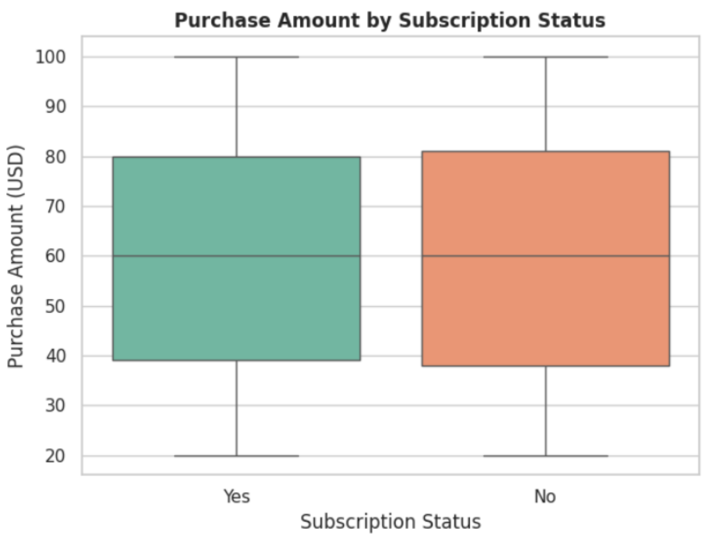


Chart 9 — Subscription Status Analysis



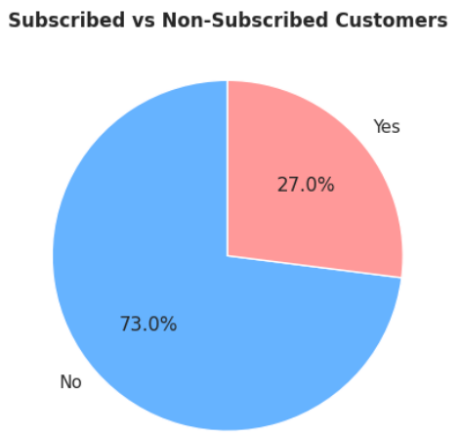
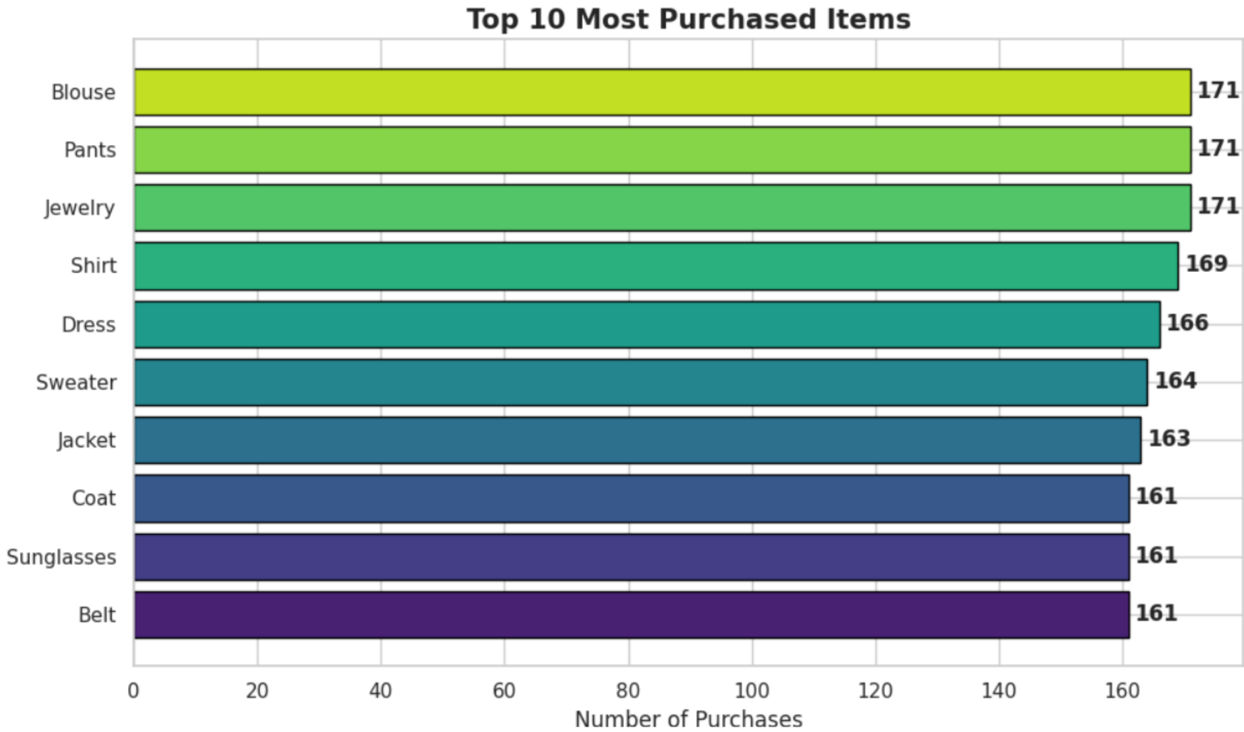


Chart 10 — Top 10 Most Purchased Items



5. Summary of Key Findings

Average Spend	Customers spend an average of ~$59.76 per transaction, with most falling in the $20–$100 range.
Top Category	Clothing is the most purchased category, driving the highest volume of transactions.
Highest Spending Group	The 36-50 age group spends the most per transaction on average.
Payment Preference	Credit Card is the most preferred payment method (dominant across all age groups).
Peak Season	Fall generates the highest total revenue; Winter follows due to holiday demand.
Best Value Category	Outerwear generates the highest average purchase value per transaction.
Subscription Impact	Subscribed customers spend slightly more, indicating higher engagement and loyalty.
Most Popular Item	Blouse, Jewelry, and Pants are the most frequently purchased individual items.

6. Business Recommendations

Based on the analysis, the following five strategic recommendations are proposed to help the business grow revenue and improve customer experience:

R 1	**Boost Clothing & Accessories Sales** Since Clothing and Accessories are the top-selling categories, dedicate the largest portion of the marketing budget to these segments. Launch seasonal campaigns specifically for Fall to capitalise on peak demand.
R 2	**Target the 36-50 Age Group with Premium Products** This age group has the highest average spend per transaction. Introduce premium product lines and personalised recommendations for this segment to maximise revenue per customer.
R 3	**Enhance Credit Card & Digital Payment Experience** With Credit Card being the most preferred payment method, ensure the checkout process is seamless and fast. Offer exclusive cashback or reward points for Credit Card users to increase retention.
R 4	**Grow the Subscription Base** Subscribed customers spend more and are more loyal. Introduce a free trial period, referral bonuses, and exclusive subscriber-only discounts to convert non-subscribed customers into subscribers.
R 5	**Optimise Inventory for Peak Season and Top Items** Stock up on the top 10 most purchased items before the Fall and Winter seasons. Feature these items on the homepage and in email campaigns to drive maximum sales during peak periods.

7. Tools and Methodology

Python 3	Core programming language used for all analysis
Pandas	Data loading, cleaning, manipulation and aggregation
NumPy	Numerical operations and array handling
Matplotlib	Base plotting library for creating all charts
Seaborn	Statistical visualisation library built on Matplotlib
Google Colab	Cloud-based Jupyter notebook environment
GitHub	Version control and project hosting platform
Kaggle	Source of the Consumer Shopping Trends dataset

Methodology

Step 1: Data Collection — Downloaded Consumer Shopping Trends dataset from Kaggle

Step 2: Data Upload — Loaded CSV file into Google Colab environment

Step 3: Data Exploration — Examined shape, columns, data types and unique values

Step 4: Data Cleaning — Checked and handled missing values and duplicates

Step 5: Feature Engineering — Created Age Group column from raw Age data

Step 6: Visualisation — Generated 10 charts covering all key business questions

Step 7: Insight Extraction — Derived findings and patterns from each chart

Step 8: Reporting — Compiled results into this professional PDF report

Step 9: Submission — Uploaded notebook and charts to GitHub repository

8. Conclusion

This project successfully demonstrated a complete end-to-end data analysis workflow — from raw data to actionable business insights. Using Python and its data science libraries, 10 visualizations were created that revealed meaningful patterns in customer behaviour across gender, age, product categories, payment methods, and seasons.

The analysis shows that the business has a strong customer base in the 26-50 age range, with Clothing being the dominant product category and Fall being the peak revenue season. Credit Card is the preferred payment method and subscribed customers demonstrate higher spending and loyalty.

The five strategic recommendations provided — targeting premium segments, optimising payment experience, growing subscriptions, and aligning inventory with seasonal demand — can help the business maximise revenue and improve customer satisfaction.

SUBMITTED BY: Garima Singh